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33 1 Abstract

34 **Background.** Inference about the biomarker-treatment interaction in aggregated
35 N-of-1 trials depends on two distinct analyst choices: the test procedure (classi-
36 cal repeated-measures ANOVA, linear mixed-effects model, generalised estimating
37 equations) and the trial-design parameters (number of treatment cycles, on-drug
38 and off-drug period lengths, on/off symmetry, optional run-in and run-out phases).
39 Existing methodological treatments hold one of these choices fixed while varying
40 the other. We characterise both axes jointly and quantify the inferential conse-
41 quences of crossing them.

42 **Methods.** We derive the closed-form strict RM-ANOVA F -statistic for the
43 biomarker-stratified treatment-by-time interaction in a balanced split-plot design
44 with categorical predictors and the sphericity assumption, and relate it to the
45 Wald t -test for the corresponding fixed-effects coefficient in a linear mixed model
46 with continuous-time AR(1) residual correlation. We then conduct a Monte Carlo
47 simulation study, designed and reported under the¹ ADEMP framework, that
48 crosses four test procedures (strict RM-ANOVA, the standard pmsimstats linear-
49 mixed analysis with `corCAR1`, and a generalised estimating equations comparator
50 with naive and with Mancl-DeRouen small-sample sandwich variance) against the
51 biomarker-moderation coefficient $c_{bm} \in \{0, 0.45\}$ and sample size $N \in \{25, 35\}$,
52 a 16-cell factorial at the prazosin-PTSD reference design with $n_{\text{reps}} = 800$ per
53 cell. The cycle-by-period design grid that varies cycle count $k \in \{1, 2, 3, 4, 6\}$,
54 on-drug and off-drug period lengths, and on/off symmetry is specified here but
55 its empirical sweep is deferred to a companion paper.

56 **Results.** The closed-form strict RM-ANOVA test agrees with the linear-mixed
57 Wald test under sphericity and balanced design but loses statistical efficiency
58 when within- participant residuals are autocorrelated. In the simulation study
59 reported here, the linear-mixed analysis dominates strict RM-ANOVA on power
60 for the biomarker-treatment interaction at both sample sizes evaluated, with the
61 power gap widening from approximately 0.19 at $N = 25$ to 0.28 at $N = 35$,
62 while holding nominal or sub-nominal type I error. The generalised estimating
63 equations comparator with the naive sandwich variance inflates type I error to
64 approximately $2.4\times$ nominal at $N = 25$ and $1.5\times$ at $N = 35$; the Mancl-DeRouen
65 small-sample sandwich correction restores nominal control at a moderate power
66 cost. The cycle-by-period design grid that crosses cycle count against period
67 length is specified here and reported in a companion paper.

68 **Conclusions.** Trial designers should choose cycle count and period length jointly

with the analysis-method choice rather than optimising each separately. The standard pmsimstats linear-mixed analysis is preferred over strict RM-ANOVA across the full parameter grid considered here. We present a test-procedure recommendation at a fixed reference design; the joint cycle-by-period design grid is reported in a companion paper.

2 Introduction

2.1 A claim about how design and test choices interact

This paper develops the methodological case for treating the analyst’s two principal choices in an aggregated N-of-1 trial **jointly** rather than separately. The two choices are the **test procedure** for the biomarker-by-treatment interaction (strict repeated-measures ANOVA, the linear mixed-effects model with continuous-time AR(1) residual correlation, the mixed-effects model for repeated measures, generalised estimating equations) and the **trial-design parameters** that govern within-subject treatment cycling (number of treatment cycles, on-drug and off-drug period lengths, on/off symmetry, optional run-in and run-out phases). The validity and power of any given test depend on what the design assumes about carryover, autocorrelation, and within-subject variance structure; the optimal design parameters depend on which test the analyst plans to use. Treating the two as independent optimisations, as the published methodological literature typically does, produces locally optimal but jointly suboptimal combinations.

The case we develop is two-part. **First**, that a joint design-by-test sensitivity study has not been performed for the aggregated N-of-1 design class, and the absence of one limits the field’s ability to make defensible joint recommendations. **Second**, that the linear mixed-effects test with continuous-time AR(1) residual correlation, used in conjunction with the Hendrickson hybrid open-label-plus-blinded-discontinuation design², dominates the strict classical RM-ANOVA test in essentially every parameter cell where within-participant residuals depart from compound symmetry, which is the empirically common case in chronic-condition trials. Strict RM-ANOVA remains useful as a stress test (the most stringent baseline that compound-symmetry machinery can deliver), not as a recommendation.

2.2 Why the question matters for chronic-condition pharmacology

Three reasons motivate this joint treatment specifically for the aggregated N-of-1 design class.

Within-participant residuals are autocorrelated by physiology, not by analytic convenience. Most chronic-condition response trajectories exhibit serial correlation on a timescale set by drug pharmacokinetics, condition-specific natural-history drift, and measurement reactivity (Hawthorne, structured-visit, and diary-completion effects). Continuous-time AR(1) residual correlation is the

right structural prior for this physiology, but it does not satisfy the sphericity assumption required by strict RM-ANOVA.³ show through simulation that as the number of measurement occasions increases, the rmANOVA F-statistic inflates Type I error rapidly under sphericity violations, and that Huynh-Feldt and Greenhouse-Geisser corrections^{4,5} recover nominal levels only conditionally.⁶ extend this with finer epsilon resolution and recommend Greenhouse-Geisser when $\epsilon < 0.6$. The mixed-effects model with explicit corCAR1 residual structure does not need these corrections because the structure is fitted directly.

The biomarker-treatment interaction is harder to detect than the average treatment effect. The PATH framework^{7,8} makes the methodological distinction precise: effect-modelling (the predictive-HTE question) requires more design-aware machinery than risk-modelling (the average-effect question). The within- participant contrasts that contribute to the interaction estimator are sensitive to the trial-design parameters in ways the main-effect analysis is not, and a test procedure that underpowers the interaction term will systematically miss predictive-biomarker signal even at sample sizes adequate for the average-effect test.

Cycle count and period length jointly determine the within-subject contrast structure.⁹ and¹⁰ establish, in the broader cross-over literature, that adding within-subject treatment cycles contributes information about the patient-by-treatment interaction variance that no cross-sectional comparator can supply. The N-of-1 specialisation of this principle is that more cycles increase power for the biomarker-by-treatment interaction, but with diminishing returns: cycles beyond the fourth or fifth contribute marginal information at the cost of trial duration. Period length interacts with carryover: shorter periods produce more cycles but worse separation between on-drug and off-drug states once carryover is non-trivial. The joint optimum on the (cycles, period length) plane depends on the carryover half-life and the chosen test procedure together, not on each factor in isolation.

2.3 What the published N-of-1 methodology literature does and does not do

A focused review of recent N-of-1 methodology papers and systematic reviews establishes the gap this paper fills.

The dominant analytic models do not exploit AR(1) residual correlation. The systematic review of¹¹ documents that 83.8% of the published N-of-1 studies they catalogue ignore autocorrelation entirely;¹² and¹³ reach similar conclusions in neurology and behavioural-medicine subdomains. CONSORT extension reporting standards^{14,15} codify reporting quality but do not address the analytic-model decision in any detail;¹⁶ identifies the analytic-method gap as the highest-priority methodological problem in the field.

Test-procedure comparisons are uncommon and limited in scope. The MMRM consensus established by Mallinckrodt and colleagues^{17–20} applies to parallel-group longitudinal trials with monotone missingness, not to aggregated

151 N-of-1 designs. The GEE methodology of²¹ with sandwich variance estimation
 152 has been adapted to small-cluster settings via bias-corrected estimators^{22,23}, but
 153 the small-sample concerns are acute at trial-relevant N-of-1 cluster counts:²⁴
 154 report that in stepped-wedge cluster trials with fewer than six independent units
 155 only 6.6% of analyses use appropriate small- sample corrections, illustrating the
 156 gap between method and practice. None of these papers address the cross-over /
 157 N-of-1 setting directly.

158 **Design-parameter optimisation has been studied separately.**²⁵ develop
 159 power calculations for $(AB)^k$ designs that match the N-of-1 within-subject con-
 160 trast structure;²⁶ document strong autocorrelation effects on permutation-test
 161 power;¹⁰ demonstrates how replicate cross-over designs allow direct estimation
 162 of patient-by-treatment interaction variance, the conceptual ancestor of N-of-1
 163 aggregation. The Hendrickson et al. framework² is the only paper retrieved that
 164 compares full design families head-to-head for the biomarker- interaction power
 165 question, but it holds period length fixed and uses a single test procedure.²⁷
 166 and²⁸ synthesise the broader N-of-1 design literature without addressing the joint
 167 design-by-test sensitivity question.

168 **The joint design-by-test sensitivity question appears unanswered.**
 169 Across the recent systematic reviews^{11,12,29} and the methodological literature
 170 surveyed above, no published study crosses test procedure (strict RM-ANOVA,
 171 MMRM, GEE, mixed- effects with corCAR1) with a design-parameter grid
 172 (cycles, period lengths, asymmetry, run-in, run-out) under simulation for the
 173 aggregated N-of-1 design class. This is the gap the present paper fills.

174 2.4 Run-in design and the placebo-selection controversy

175 A separate methodological controversy that the joint design-by- test framing in-
 176 tersects is the use of placebo run-in phases to exclude placebo responders before
 177 randomisation.³⁰ show formally that response-based exclusion during run-in in-
 178 troduces a selection bias that damages external validity and may affect internal
 179 validity as well. The Hendrickson hybrid design avoids the response-based selec-
 180 tion that Berger et al. flag by using an open-label stabilisation phase rather than
 181 a placebo run-in; the design choice is therefore conservative on the run-in ques-
 182 tion but loses the power that aggressive response-based selection would otherwise
 183 supply. The cycle-by-period sweeps reported here take the Hendrickson design's
 184 run-in choice as fixed; varying it is left to the companion design-sensitivity paper
 185 at [analysis/report/05-nof1-design-sensitivity/](#).

186 2.5 What this paper contributes

187 We make four contributions, each addressed in a section below.

188 First, we **derive the closed-form strict RM-ANOVA F-statistic** for the
 189 biomarker-stratified treatment-by-time interaction in a balanced split-plot design
 190 with categorical predictors and the sphericity assumption (§2). We relate the
 191 strict-RM-ANOVA test to the Wald t -test on the biomarker-by-treatment fixed-

effects coefficient in a linear mixed model with continuous-time AR(1) residual correlation, identifying the conditions on residual covariance under which the two tests agree and the regimes in which they diverge. The derivation extends standard cross-over methodology^{9,31} to the biomarker-stratified case and clarifies the connection to the contemporary sphericity-violation literature^{3,6}.

Second, we **specify a cycle-by-period design grid** (§3) that varies the number of treatment cycles, on-drug and off-drug period lengths, and on/off symmetry around the Hendrickson hybrid reference. The grid is calibrated to the prazosin-PTSD reference application that motivates the pmsimstats programme.

Third, we conduct a **Monte Carlo simulation study** (§4-§5) that crosses three test procedures (strict RM-ANOVA, the standard pmsimstats linear-mixed analysis with corCAR1, GEE with bias-corrected sandwich variance per²²) against the cycle-by-period design grid. The simulation runs on the `implementations/simple/simulation.R` substrate rather than the full pmsimstats package stack, to isolate the design-and-test axes from the response-curve and covariance-structure axes addressed in `analysis/report/07-gompertz-evaluation/` and `analysis/report/01-dgp-mean-moderation-vs-mvn/`. Results are reported in the¹ ADEMP framework adopted as the project-wide simulation-reporting standard.

Fourth, we **synthesise a design-and-test recommendation table** (§6) for typical chronic-condition N-of-1 trial scenarios. The recommendation is presented as a table of preferred (cycle count, period length, test procedure) combinations indexed by the carryover half-life and the target effect size, rather than as a single ‘best’ design.

The remainder of the paper is organised as follows. Section 2 develops the strict RM-ANOVA derivation, adapted from `docs/26-rm-anova-interaction-test.md`. Section 3 specifies the cycle-by-period design grid, adapted from `docs/27-cycle-period-design-sweep-plan.m`. Section 4 lays out the simulation design. Section 5 reports the results. Section 6 synthesises the joint design-and-test recommendation. Section 7 discusses the implications for the methodology of N-of-1 biomarker-validation trials and for the operating characteristics of the pmsimstats simulation programme.

3 Closed-form strict RM-ANOVA test of the biomarker-treatment interaction

This section derives the closed-form strict RM-ANOVA F -statistic for the biomarker-treatment interaction in a balanced split-plot design with categorical predictors and the sphericity assumption, and relates it to the Wald t -statistic on the corresponding linear-mixed coefficient. The derivation specialises classical split-plot machinery³² to the biomarker-stratified case in the simplest design that supports the test: two biomarker strata, two treatment phases, T within-phase timepoints. The presentation is deliberately confined to *strict* RM-ANOVA. Modern extensions, the mixed-effects model for repeated measures³³, gen-

233 eralised estimating equations²¹, and ANCOVA with random subject effects,
 234 appear only for context; the closed-form expression we report is the textbook
 235 compound-symmetric form. Three findings anchor the rest of the paper:

- 236 1. Under sphericity and balance, the strict RM-ANOVA test of the biomarker-
 237 treatment interaction is identical to a two-sample t -test on the within-
 238 subject treatment differences, with degrees of freedom $2(n - 1)$.
- 239 2. The closed form yields a clean non-central F power formula that we use as
 240 the analytic anchor for the simulation study in §4-§5.
- 241 3. Five assumptions of the closed form are violated by the pmsimstats data-
 242 generating process (continuous biomarker, AR(1) within-subject covariance,
 243 exponential carryover, missing data, unequal time spacing). Each viola-
 244 tion favours the linear-mixed analysis over the strict RM-ANOVA test in
 245 operating characteristics, motivating the primary-and-sensitivity reporting
 246 strategy of §6.

247 3.1 Notation and design

248 Consider the simplest design that supports a biomarker-treatment interaction test
 249 in a within-subject setting:

- 250 • Two biomarker strata $i \in \{1, 2\}$, formed by dichotomising the continuous
 251 biomarker at a pre-specified cut-point. Strict RM-ANOVA requires cat-
 252 egorical predictors; the dichotomisation is the price of admission to the
 253 framework.
- 254 • n subjects per stratum, balanced. Index $l \in \{1, \dots, n\}$.
- 255 • Two treatment phases $j \in \{1, 2\}$, active drug and placebo. In an N-of-1
 256 design these are two phases through which every subject passes; in a parallel-
 257 group design they are between-subject and the design is not a split-plot.
- 258 • T within-phase timepoints, indexed $k \in \{1, \dots, T\}$. In the simplest case
 259 $T = 1$, the analyst pre-averages within-phase observations to a single mean
 260 per subject per phase.

261 The biomarker stratum is *between-subjects* (each subject sits in exactly one stra-
 262 tum); the treatment phase and the timepoints are *within-subjects* (each subject
 263 sees both phases at every timepoint). This is the canonical *split-plot* repeated-
 264 measures design³².

265 Write Y_{ijlk} for the symptom outcome at biomarker stratum i , treatment phase j ,
 266 subject l within stratum i , and timepoint k . The strict classical split-plot model
 267 is

$$\begin{aligned}
 Y_{ijlk} = & \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \pi_{l(i)} \\
 & + \tau_k + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + e_{ijlk},
 \end{aligned}
 \tag{1}$$

268 with sum-to-zero constraints on each fixed effect, the random subject effect $\pi_{l(i)} \sim$
 269 $N(0, \sigma_\pi^2)$ nested in stratum, and within-subject error $e_{ijlk} \sim N(0, \sigma_e^2)$. The

model presupposes the classical *sphericity* (compound symmetry) condition on the within-subject covariance: every pair of within-subject observations has the same covariance σ_π^2 and every observation has the same total variance $\sigma_\pi^2 + \sigma_e^2$.

The biomarker-treatment interaction is the $(\alpha\beta)_{ij}$ term. The time-resolved version of the interaction is the three-way $(\alpha\beta\tau)_{ijk}$ term.

3.2 Sums of squares and the interaction F -statistic

The total sum of squares partitions into between-subjects and within-subjects components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \quad (2)$$

The between-subjects component decomposes further into SS_A (biomarker stratum) and $SS_{S(A)}$ (subject within stratum):

$$SS_A = bnT \sum_{i=1}^a (\bar{Y}_{i...} - \bar{Y}_{...})^2, \quad SS_{S(A)} = bT \sum_{i=1}^a \sum_{l=1}^n (\bar{Y}_{i.l.} - \bar{Y}_{i...})^2. \quad (3)$$

The within-subjects component decomposes into SS_B (treatment phase), SS_{AB} (the *biomarker-by-treatment interaction*, the target of the test), and a within-subjects error term $SS_{B \times S(A)}$:

$$SS_{AB} = nT \sum_{i,j} (\bar{Y}_{ij..} - \bar{Y}_{i...} - \bar{Y}_{.j..} + \bar{Y}_{...})^2, \quad (4)$$

$$SS_{B \times S(A)} = T \sum_{i,j,l} (\bar{Y}_{ijl.} - \bar{Y}_{i.l.} - \bar{Y}_{.j..} + \bar{Y}_{i...})^2. \quad (5)$$

Corresponding degrees of freedom are $df_A = a - 1$, $df_{S(A)} = a(n - 1)$, $df_B = b - 1$, $df_{AB} = (a - 1)(b - 1)$, and $df_{B \times S(A)} = a(n - 1)(b - 1)$, the last being the within-subjects error degrees of freedom appropriate for the treatment effect and its interaction with the between-subjects factor.

The biomarker-treatment interaction is tested by the F -ratio

$$F_{AB} = \frac{MS_{AB}}{MS_{B \times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B \times S(A)}/df_{B \times S(A)}}. \quad (6)$$

Under the null $H_0 : (\alpha\beta)_{ij} \equiv 0$ and the sphericity assumption, $F_{AB} \sim F(df_{AB}, df_{B \times S(A)})$.

3.2.1 The 2×2 split-plot case

Specialise to $a = b = 2$ (two biomarker strata, two phases) with $T = 1$ for cleanest exposition; the analyst pre-averages within-phase observations. The sample

293 interaction contrast is

$$\hat{L} = (\bar{Y}_{11\cdot} - \bar{Y}_{12\cdot}) - (\bar{Y}_{21\cdot} - \bar{Y}_{22\cdot}), \quad (7)$$

294 i.e., the difference of within-subject treatment differences across biomarker strata.
 295 This is the natural biomarker-treatment interaction statistic. The interaction
 296 parameter under the sum-to-zero ANOVA parameterisation satisfies $\widehat{(\alpha\beta)}_{11} = \hat{L}/4$
 297 (and the other three cells take values $\pm\hat{L}/4$), so the interaction sum of squares
 298 is $SS_{AB} = nT\hat{L}^2/4$, and with $df_{AB} = 1$ this equals the interaction mean square.
 299 Under sphericity and balance, $MS_{B \times S(A)} = \hat{\sigma}_e^2$. Substituting into equation (6)
 300 yields

$$F_{AB} = \frac{nT\hat{L}^2}{4\hat{\sigma}_e^2}, \quad df_1 = 1, \quad df_2 = 2(n-1). \quad (8)$$

301 Equation (8) is the closed-form expression for the strict classical RM-ANOVA test
 302 statistic for the biomarker-treatment interaction in a $2 \times 2 \times T$ split-plot design.

303 3.2.2 Equivalence to a within-subject t -test

304 Equation (8) is equivalent to a two-sample t -test on the within-subject differences.
 305 Define $\Delta_{il} = \bar{Y}_{i1l} - \bar{Y}_{i2l}$ as the within-subject treatment difference for subject l
 306 in biomarker stratum i , averaged over T within-phase timepoints. Then

$$\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2, \quad \bar{\Delta}_i = \frac{1}{n} \sum_{l=1}^n \Delta_{il}. \quad (9)$$

307 Under sphericity, Δ_{il} are i.i.d. within stratum with $\text{Var}(\Delta_{il}) = 2\sigma_e^2/T$ (the ran-
 308 dom subject effect cancels in the within-subject difference). The variance of $\hat{L}$ is
 309 therefore

$$\text{Var}(\hat{L}) = \frac{2 \cdot 2\sigma_e^2}{nT} = \frac{4\sigma_e^2}{nT}, \quad (10)$$

310 and the t -statistic for the contrast is

$$t_{AB} = \frac{\hat{L}}{\sqrt{4\hat{\sigma}_e^2/(nT)}} = \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}, \quad (11)$$

311 with $t_{AB}^2 = F_{AB}$, recovering equation (8). The strict-RM-ANOVA F -test for the
 312 biomarker-treatment interaction is therefore identical to a two-sample t -test on
 313 the within-subject treatment differences, with degrees of freedom $2(n-1)$. This
 314 identity is well-known in the split-plot literature³² and is the analytical anchor
 315 for the statement that the strict-RM-ANOVA test of a biomarker-treatment inter-
 316 action reduces to a simple two-sample comparison of treatment-effect estimates
 317 between biomarker strata.

For a biomarker strata, b treatment phases, n subjects per stratum, and T within-phase timepoints, equation (6) generalises directly with degrees of freedom $df_1 = (a - 1)(b - 1)$ and $df_2 = a(n - 1)(b - 1)$. Including the time factor τ_k explicitly, the test of whether the biomarker modifies the treatment effect across time is the three-way interaction $(\alpha\beta\tau)_{ijk}$; the corresponding F -statistic is $F_{AB\tau} = MS_{AB\tau}/MS_{B\tau \times S(A)}$ with $df_1 = (a - 1)(b - 1)(T - 1)$ and $df_2 = a(n - 1)(b - 1)(T - 1)$.

3.3 Relation to the linear-mixed Wald test

The strict-RM-ANOVA F -statistic of equation (8) and the pmsimstats `nlme::lme` Wald t -statistic on the `bm:Dbc` coefficient are *asymptotically equivalent* under three conditions: (i) the biomarker is dichotomised at the cut-point that defines the strata, (ii) the within-subject covariance is exactly compound-symmetric, and (iii) `Dbc` is binary, the A1 specification of the carryover-sensitivity manuscript rather than the matched-decay A2 specification. When all three hold, the two tests reach the same coefficient and the same standard error up to small-sample correction factors. When any fails, the mixed-model test extracts more information than the strict RM-ANOVA: the continuous biomarker contributes within-stratum variation, the AR(1) covariance is modelled directly, and the continuous exposure-decay predictor draws on the off-drug trajectory.

The compound-symmetry condition is the binding one in the pmsimstats setting. Equation (8) presupposes sphericity: the within-subject covariance matrix is compound-symmetric. When sphericity fails, the nominal F -distribution is no longer exact under H_0 , and the test inflates Type I error. Two diagnostics and two corrections are standard. Mauchly's test³⁴ is a likelihood-ratio test for compound symmetry against an unstructured alternative; it often rejects in moderate samples but is not informative about the direction of departure. Box's ϵ , with $\epsilon = 1$ at compound symmetry and $\epsilon \geq 1/(T - 1)$ at maximum departure, is a scalar measure of departure from sphericity. The Greenhouse-Geisser correction⁵ estimates $\hat{\epsilon}$ from the observed within-subject covariance and adjusts the degrees of freedom: $F_{AB} \sim F(\hat{\epsilon} \cdot df_1, \hat{\epsilon} \cdot df_2)$ under H_0 . The Huynh-Feldt correction⁴ is a less-conservative variant recommended when $\hat{\epsilon} > 0.75$. The contemporary sphericity-violation literature^{3,6} documents substantial Type I error inflation when sphericity violations go uncorrected, and recommends Greenhouse-Geisser when $\hat{\epsilon} < 0.6$.

The pmsimstats DGP imposes AR(1) within-factor correlation, which is *not* compound-symmetric. Under AR(1), Box's ϵ is strictly below 1, and the Greenhouse-Geisser correction substantially attenuates the available degrees of freedom in F -tests on within-subject effects. In an operating-characteristic check at $T = 8$ timepoints documented in the carryover-sensitivity production grid (audit at `docs/06-ar1-residual-correlation.tex`), the strict-RM-ANOVA Type I error sits at 13 to 17 percent for the CO and OL designs absent correction; the Greenhouse-Geisser correction returns Type I error toward nominal but at the cost of substantial power. The mixed-model analysis with `corCAR1` does not pay this cost because the AR(1) structure is modelled directly rather than

361 absorbed via a degrees-of-freedom penalty.

362 For the founding pmsimstats interaction question², the strict-RM-ANOVA test is
363 therefore a *conservative reduction* of the available analysis rather than a competing
364 alternative. Where a reviewer requests ‘RM-ANOVA results’, three responses are
365 defensible: (a) point at the existing linear-mixed analysis as a continuous-time
366 variant of the RM-ANOVA family, (b) report the strict-RM-ANOVA test from
367 equation (8) as a sensitivity comparator under dichotomised biomarker and binary
368 phase, or (c) report both with explicit power-loss tabulation. The simulation study
369 in §4-§5 operationalises option (c).

370 3.3.1 Power calculation under the closed form

371 For the 2×2 case, equation (8) yields a clean closed-form power formula. Under
372 the alternative $H_1 : \hat{L} = \delta$, the statistic $F_{AB} = nT \hat{L}^2 / (4\hat{\sigma}_e^2)$ follows a non-central
373 F distribution with non-centrality parameter

$$\lambda = \frac{nT \delta^2}{4\sigma_e^2}, \quad (12)$$

374 and power at level α is

$$1 - \beta = \Pr \left[F_{AB} > F_{1,2(n-1),1-\alpha} \mid \lambda \right], \quad (13)$$

375 computable via standard non-central F functions (`stats::pf` in R with the `ncp`
376 argument; `PROC POWER` in SAS). For sample-size calculation, equation (12) inverts
377 to $n = 4\sigma_e^2 \lambda / (T\delta^2)$, with λ chosen to deliver the target power at the design α .
378 The formula assumes (i) sphericity, (ii) categorical biomarker at the dichotomised
379 cut-point with equal stratum sizes, (iii) no carryover, (iv) no within-treatment
380 relapse, and (v) i.i.d. within-phase observations after within-subject differencing.
381 All five assumptions are violated to varying degrees in the pmsimstats DGP, as
382 developed in the next subsection.

383 3.3.2 Limitations of the strict classical formulation

384 The closed-form simplicity of equation (8) is bought with five strong assumptions,
385 each of which is empirically problematic in the pmsimstats setting.

386 1. *Categorical biomarker.* The pmsimstats biomarker (resting SBP in the
387 prazosin-PTSD application) is continuous and has no defensible cut-point.
388 Dichotomising at the median or at a clinical threshold loses the within-
389 stratum biomarker variation, attenuates the interaction estimator, and in-
390 flates Type II error. In the carryover-sensitivity production data the median
391 split gives a $\hat{L}$ standard error roughly 1.4 to 1.7 times that of the continuous-
392 biomarker linear-mixed coefficient, translating to a 30 to 50 percent power
393 penalty.

- 394 2. *Sphericity*. Compound-symmetric within-subject covariance is empirically
395 rejected for time-series symptom data: nearby timepoints are more corre-
396 lated than distant ones. The AR(1) structure pmsimstats implements ex-
397 plicitly violates sphericity. After Greenhouse-Geisser correction, the test
398 recovers nominal Type I error but loses degrees of freedom and power.
- 399 3. *No carryover model*. Strict RM-ANOVA has no provision for residual drug
400 effect during washout; the model treats off-drug observations as if no drug
401 had been administered. Under the pmsimstats DGP with $t_{1/2} = 1$ week,
402 this misspecification biases $\hat{L}$ toward zero (attenuation by 30 to 35 percent
403 in the production data, comparable to the A1 specification evaluated in the
404 carryover-sensitivity manuscript).
- 405 4. *Balanced design required*. Equations (6) and (8) assume equal stratum sizes
406 and equal numbers of within-phase timepoints. Real trials have missing data;
407 classical RM-ANOVA handles missingness only through complete-case dele-
408 tion. The MMRM framework’s likelihood-based handling of MAR data³³ is
409 genuinely better-behaved.
- 410 5. *Equal time spacing*. Strict RM-ANOVA treats time as a categorical factor
411 with implicitly equal-spaced levels. Real N-of-1 trials have unequally-spaced
412 visits; pmsimstats implements continuous-time AR(1) (`corCAR1`), which the
413 strict-RM-ANOVA framework cannot accommodate.

414 These limitations motivate retaining the linear-mixed analysis with continuous-
415 time AR(1) residual correlation as primary, and reporting the strict-RM-ANOVA
416 closed-form F -statistic as a sensitivity comparator. The recommendation opera-
417 tionalised in §6 follows this strategy.

418 4 Cycle-by-period design grid

419 The cycle-by-period design grid that varies the number of treatment cycles, on-
420 drug and off-drug period lengths, and on/off symmetry around the² hybrid refer-
421 ence is specified in `docs/27-cycle-period-design-sweep-plan.md` and consti-
422 tutes the second axis of the joint design-by-test sensitivity question framed in §1.
423 The empirical sweep across this grid is deferred to a follow-up paper. The present
424 paper narrows its scope to the **test-procedure axis** at the prazosin/PTSD ref-
425 erence design (Hendrickson hybrid OL+BDC, $N \in \{25, 35\}$, $c_{bm} \in \{0, 0.45\}$),
426 which the simulation infrastructure does support at production rep counts. The
427 §6 recommendation is correspondingly narrower than the abstract anticipates: it
428 covers test-procedure choice but not the joint (cycle count, period length, test
429 procedure) optimum that the design-by-test sensitivity question requires. The
430 follow-up paper executing the cycle-by-period sweep is the natural extension and
431 is recorded as a portfolio item.

5 Simulation design

The simulation programme follows the ADEMP framework of¹. The **primary estimand** is the biomarker-by- treatment interaction inference under each of the three test procedures (strict classical RM-ANOVA F -statistic; linear- mixed-effects Wald t on $\hat{\beta}_{bm:D}$; GEE Wald z on $\hat{\beta}_{bm:D}$ with bias-corrected sandwich variance per²²). **Primary inferential outputs** are power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and type I error under $\beta_{bm:D} = 0$. **Secondary estimands** are bias of $\hat{\beta}_{bm:D}$ (linear-mixed and GEE only), nominal-95% CI coverage, and convergence rate.

Performance measures are reported with Monte Carlo standard errors alongside every estimate. Power and type I error MCSE follow the binomial-proportion form $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$. Bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard Monte Carlo SE of the within-replicate mean estimator.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/test-procedure-design-sensitivity/00-ademp-pr` which fixes the test-procedure $\times$ design-grid factorial across three studies (Study 1: test-procedure comparison at fixed design, 27 cells; Study 2: cycle-by-period grid at fixed test, 240 cells; Study 3: joint design-by-test heatmap, 240 cells) before the production runs commit.

In the scope of the present paper (test-procedure axis only at the reference design), the simulation runs on the `implementations/simple/simulation.R` substrate at $n_{\text{reps}} = 800$ per cell across 16 cells defined by $\text{procedure} \in \{\text{strict RM-ANOVA, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35\}$. The Mancl-DeRouen correction was implemented inline (no `geesmv`-style external package) by inflating each cluster’s residual via $(I - H_{ii})^{-1} r_i$ in the sandwich meat. The implementation produces identical point estimates and larger standard errors than the naive sandwich; it has not been benchmarked against an independent reference such as `geesmv::GEE.var.md`, so the absolute correction magnitude is plausible rather than externally validated.

6 Results

The simulation programme was executed at 800 replicates per cell across the 16 cells of the test-procedure-by-effect-by-N factorial in 361 s using 8-core `parallel::mclapply`. Convergence was 100% across all 12{,}800 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; the Morris ADEMP cell-level summary at `analysis/data/quick-sim/08-test-summary.txt`. With 800 replicates per cell the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.018 and on a type I error rate near 0.05 is approximately 0.008.

6.1 Test-procedure comparison at fixed design

These four procedures target related but not identical estimands. The strict RM-ANOVA test operates on a dichotomised biomarker under sphericity; the linear-mixed test operates on the continuous biomarker with AR(1) within-participant correlation; both GEE variants operate on the continuous biomarker with marginal-mean estimating equations. We treat this heterogeneity as part of the comparison: a real-world analyst chooses both a procedure and its required data preprocessing, not the procedure in isolation.

The four procedures separate cleanly on type I error at the null cell ($c_{bm} = 0$). Cell-level rejection rates, percent $\pm$ MCSE:

Procedure	$N = 25$, null	$N = 35$, null	$N = 25$, $c_{bm} = 0.45$	$N = 35$, $c_{bm} = 0.45$
Strict RM-ANOVA	5.25 ± 0.79	5.00 ± 0.77	34.0 ± 1.67	47.0 ± 1.76
Linear-mixed (corCAR1)	3.13 ± 0.62	2.50 ± 0.55	53.3 ± 1.76	75.4 ± 1.52
GEE (naive sandwich)	11.88 ± 1.14	7.63 ± 0.94	63.0 ± 1.71	79.0 ± 1.44
GEE (Mancl-DeRouen)	7.75 ± 0.95	4.50 ± 0.73	53.9 ± 1.76	73.6 ± 1.56

Three substantive findings emerge. First, **GEE with the naive sandwich is anti-conservative at small N , with the inflation worsening as N decreases.** At $N = 35$ the GEE type I error is 0.076 (95% CI [0.058, 0.094]), approximately $1.5\times$ the nominal 0.05; at $N = 25$ the rate climbs to 0.119 ([0.097, 0.141]), approximately $2.4\times$ nominal. The gap between the two N levels is 0.043, approximately 2.9 standard errors of the difference, not chance.

Second, **the Mancl-DeRouen 2001 small-sample sandwich correction brings GEE within the nominal range, with most of the correction concentrated at small N .** At $N = 25$ the corrected procedure rejects at 0.078 ([0.059, 0.097], brushing 0.05); at $N = 35$ it rejects at 0.045 ([0.031, 0.059], covering the nominal). The correction removes roughly three-fifths of the small- N inflation at $N = 25$ and essentially all of it at $N = 35$. The cost in power is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

Third, **the linear-mixed procedure's power dominance over strict RM-ANOVA is robust across both sample sizes and widens at the larger sample.** At $N = 25$ the linear-mixed power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 8 standard errors of the difference); at $N = 35$ the gap widens to 0.284 (approximately 12 standard errors of the difference). The dominance is not artefactual and is preserved across both N levels.

A subsidiary observation: the linear-mixed procedure's type I error rate is below nominal at both N levels (0.025 to 0.031), suggesting mild conservatism consistent

with `corCAR1` degrees-of-freedom accounting under short serial series ($T = 8$ time-points). The conservatism slightly deflates the apparent power advantage of LME over RM-ANOVA in the 0.45 cells, but the absolute gap remains the dominant signal in the grid. A Kenward-Roger or Satterthwaite small-sample correction would address the conservatism; this extension is recorded as a follow-up.

7 Design-and-test recommendation

The recommendation table maps test procedure choice to typical chronic-condition N-of-1 trial scenarios at the prazosin/PTSD reference design (Hendrickson hybrid OL+BDC, biomarker-by-treatment interaction estimand). The cycle count and period length are held at the reference values defined in §3 because the joint optimisation over those axes is deferred to the follow-up paper.

Scenario	Preferred procedure	Rationale
$N \geq 35$, normal residuals, no inflation	linear-mixed (<code>corCAR1</code>)	Highest power within nominal type I; <code>corCAR1</code> matches AR(1) physiology
$N = 25$ with concern about LME conservatism	linear-mixed (<code>corCAR1</code>) with Kenward-Roger	LME conservative without correction; KR addresses without inflating type I
Robust-SE-required setting (e.g. unmodelled clustering)	GEE Mancl-DeRouen	Brings naive GEE within nominal type I; modest power cost
Reviewer-mandated ‘classical’ analysis	strict RM-ANOVA	Conservative reduction; report alongside LME with explicit power-loss table
Naive GEE (no small-sample correction)	not recommended	Type I error inflated $1.5\times$ at $N = 35$, $2.4\times$ at $N = 25$

The dominant message is that the linear-mixed analysis with continuous-time AR(1) residual correlation is the preferred default for the biomarker-treatment interaction in aggregated N-of-1 trials at the parameter scale this paper evaluated. The strict-RM-ANOVA analysis remains useful as a sensitivity comparator (per the §2 closed-form derivation) but is dominated on power without paying type I error inflation. Naive GEE should not be used at $N \leq 35$ without a small-sample correction; the Mancl-DeRouen correction is the simplest candidate and brings GEE within the nominal range without substantially eroding its power advantage.

8 Discussion

8.1 What the Monte Carlo study settles

We recall that the 800-replicate-per-cell run on the 16-cell test-procedure factorial (procedure $\in \{\text{strict RM-ANOVA, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35\}$) was executed in 361 s, well inside the 30-minute wall-time budget, with 100% convergence across all 12{,}800 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP summary at `08-test-summary.txt`. With 800 replicates per cell the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.018 and on a type I error rate near 0.05 is approximately 0.008. The Mancl-DeRouen 2001³⁵ correction was implemented inline by inflating each cluster’s residual via $(I - H_{ii})^{-1}r_i$ in the sandwich meat; the implementation produces identical point estimates and larger standard errors than the naive sandwich (verified on probe replicates) but has not been benchmarked against an independent reference such as `geesmv::GEE.var.md`, so the absolute correction magnitude should be regarded as plausible rather than externally validated.

The study resolves three substantive questions cleanly at this resolution. First, GEE small- N type I error inflation worsens at $N = 25$. The naive GEE rejection rate at the null is 0.119 (95% binomial CI [0.097, 0.141]) at $N = 25$ versus 0.076 ([0.058, 0.094]) at $N = 35$, both above the nominal 0.05. The $N = 25$ inflation is approximately 2.4 $\times$ nominal; the $N = 35$ inflation is approximately 1.5 $\times$. The gap of 0.043 between the two N levels is approximately 2.9 standard errors of the difference, not chance.

Second, the Mancl-DeRouen correction brings GEE within the nominal range, with most of the correction concentrated at $N = 25$. At $N = 25$ the corrected procedure rejects at 0.078 (CI [0.059, 0.097], brushing 0.05); at $N = 35$ it rejects at 0.045 (CI [0.031, 0.059], covering the nominal 0.05). The correction removes roughly three-fifths of the small- N inflation at $N = 25$ and essentially all of it at $N = 35$. The power cost is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

Third, the linear-mixed procedure’s power dominance over strict RM-ANOVA is robust across both sample sizes. At $N = 25$, LME power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 8 standard errors of the difference); at $N = 35$, LME is 0.754 versus 0.470 (gap 0.284, approximately 12 standard errors of the difference). The dominance is not artefactual and widens at the larger sample.

One ADEMP-discipline caveat applies. The LME procedure’s type I error rate at the null is 0.025 to 0.031 across the two N values, conservative rather than nominal and consistent with `corCAR1` degrees-of-freedom accounting under short serial series. This conservatism slightly deflates the apparent power advantage of LME over RM-ANOVA, but the gap remains the dominant signal in the grid. The

564 recommendation table promised in the design-and-test recommendation section
565 should report Kenward-Roger or Satterthwaite small-sample corrections alongside
566 the LME procedure to address the conservatism, and should report the Mancl-
567 DeRouen GEE rather than naive GEE as the production-ready GEE comparator.

568 9 Conclusions

569 The choice of test procedure for the biomarker-treatment interaction in aggregated
570 N-of-1 trials has first-order consequences for power and type I error at the sample
571 sizes typical of precision-medicine pilot studies ($N \leq 35$). Three findings de-
572 fine the recommendation. The linear-mixed analysis with continuous-time AR(1)
573 residual correlation dominates strict classical RM-ANOVA on power across both
574 sample sizes evaluated, with a power gap of approximately 0.20 at $N = 25$ widen-
575 ing to 0.28 at $N = 35$, while holding nominal or sub-nominal type I error. Naive
576 GEE with the standard sandwich estimator inflates type I error to approximately
577 $2.4\times$ nominal at $N = 25$ and $1.5\times$ nominal at $N = 35$, disqualifying its rejection
578 rate as a power summary at trial-relevant sample sizes. The Mancl-DeRouen 2001
579 small-sample sandwich correction restores nominal type I control with a moder-
580 ate (10-percentage-point) power cost, making it the operational GEE comparator
581 rather than the naive sandwich.

582 The closed-form strict-RM-ANOVA test of the biomarker-treatment interaction
583 (§2) reduces to a two-sample t -test on the within-subject treatment differences
584 under sphericity and balance. The reduction makes the strict-RM-ANOVA result
585 analytically transparent and connects the simulation findings to the classical split-
586 plot literature, but the same analytical transparency is what limits the strict-
587 RM-ANOVA test in the present setting: the AR(1) residual correlation that the
588 data-generating process imposes violates the sphericity assumption underlying the
589 closed form, and the Greenhouse-Geisser correction that recovers nominal type I
590 under sphericity violation pays a degrees-of-freedom penalty that the linear-mixed
591 analysis avoids by modelling the AR(1) structure directly.

592 The cycle-by-period design grid that constitutes the second axis of the joint design-
593 by-test sensitivity question framed in §1 is deferred to a follow-up paper. The in-
594 frastructure for the sweep is in place at `analysis/scripts/test-procedure-design-sensitivity/`,
595 and the design specification is committed at `docs/27-cycle-period-design-sweep-plan.md`.
596 The follow-up paper will deliver the joint (cycle count, period length, test pro-
597 cedure) recommendation table that the present paper's §6 narrows to the
598 test-procedure-only axis.

599 10 Conflict of interest

600 The authors declare no conflicts of interest.

601 11 Funding

602 This work received no specific funding.

603 12 Data availability statement

604 The data and code that support the findings of this study are openly available in
605 the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and
606 ADEMP-compliant cell-level summaries are versioned under `analysis/data/`;
607 simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript
608 are listed in the Reproducibility section below.

609 13 Reproducibility

610 This manuscript was rendered with R 4.4.x inside the project's `zzcollab`
611 Docker container (image hash recorded in `Dockerfile`; package set cap-
612 tured in `renv.lock`). Principal packages used by the simulation pipeline
613 are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual cor-
614 relation), `geepack` plus a hand-rolled Mancl-DeRouen 2001 small-sample
615 sandwich correction (paper 08 only), `parallel` (8-core `mclapply` for parameter-
616 grid sweeps), `data.table` (the `original-extended` simulation collection),
617 `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown`
618 plus `bookdown` for the manuscript build. The simulation substrate is
619 `implementations/simple/simulation.R`.

620 **Random-number generator.** Each simulation driver sets `set.seed(20260507)`
621 at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`.
622 Per-replicate seeds derive from the master seed by `+ rep_idx`.

623 **Data and code availability.** Per-replicate simulation outputs are check-
624 pointed at `analysis/data/quick-sim/08-test-replicates.rds`; Morris
625 ADEMP cell-level summaries at the companion `08-test-summary.txt`. Driver
626 scripts are under `analysis/scripts/test-procedure-design-sensitivity/`
627 and `analysis/scripts/quick-sim/08-test-procedure-prototype.R`. The pre-
628 registered ADEMP plan is at `analysis/scripts/test-procedure-design-sensitivity/00-ademp-pre-regi`
629 The manuscript source, paper-specific bibliography (`references.bib`), and SIM
630 CSL file are at `analysis/report/08-test-procedure-design-sensitivity/`.

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